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Key Concepts and Intuition

AIML Interview Preparation

In Progress



Reading Material

⌚ 2 min read

Why does vectorization actually work faster? NumPy arrays are stored in contiguous memory blocks. Operations are delegated to pre-compiled C/Fortran routines that process chunks of data with CPU-level parallelism (SIMD instructions). Python loops, by contrast, have per-iteration interpreter overhead.

When does broadcasting fail? When two dimensions are neither equal nor one of them is 1. Example: shapes (3,4) and (3,5) cannot broadcast along the second axis — they are incompatible.

View vs. Copy — a common source of bugs: Slicing creates a view; boolean indexing creates a copy. If you modify a slice, the original changes. If you modify a boolean-indexed result, the original is unaffected.

```
arr = np.array([1,2,3,4,5])
view = arr[1:4]
view[0] = 99          # arr is now [1, 99, 3, 4, 5]

copy = arr[arr > 2]
copy[0] = 0           # arr is unchanged
```

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